

IP Workshop Manual

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- GitHub link for the IP-Workshop Repository:
<https://github.com/NinadWalke/ip-workshop>

Commands

1. Create React App: `npx create-react-app frontend`
2. Create Vite React: `npm create vite@latest`
3. Install Axios for Frontend: `npm install axios`
4. Install Backend Dependencies: `npm install express cors`
5. Install nodemon: `npm i -g nodemon`

Links

1. NodeJS: <https://nodejs.org/en/download>
2. VSCode: <https://code.visualstudio.com/docs/?dv=win64user>

Theory

1. Components: Help to divide code into files so the code can be easily reused
2. Props: Data passed to components to display it
3. Hooks: Used to manage state on the web and perform actions if the state of a component changes [**useState**, **useEffect**, *useRef*, *useMemo*]
4. API: Application Programming Interface, used to communicate and exchange data between different apps and frameworks.
5. NodeJS: What is it and why?
6. ExpressJS: What is it and why?
7. Request types (REST APIs: GET, POST, PUT, DELETE)

React Flow

1. Explain React first
2. Explain React Components, their usage and why it's better than HTML
3. Explain React Props, how they are used to pass data and why (relate to components)
4. Hooks [**useState**, **useEffect**]
5. What are APIs, just the basics